OpenAPI Contract Specification Diff & Comparison

Analysis of main branch (newcp.new.openapi.yaml) vs canonical contract (newcp.openapi.yaml)

Base Contract (Main Branch): openapi/newcp.new.openapi.yaml	**Target Contract (Canonical):** openapi/newcp.openapi.yaml
Branch Scope: main vs hotfix/adjust-api-contract	**Analysis Date:** October 2026

EXECUTIVE PURPOSE OF THE CONTRACT ADJUSTMENTS:
The initial OpenAPI contract on the main branch (referenced as newcp.new.openapi.yaml) represented a transitional draft that retained dual compatibility modes: it accepted clean polymorphic answers alongside five legacy carrier fields, while leaving 16 core PostgreSQL database columns optional in ServiceItem.required.
The refined canonical contract (newcp.openapi.yaml) eliminates dual-mode ambiguities by enforcing **strict schema validation**, **purging all 5 legacy carrier properties**, and **expanding ServiceItem required fields from 8 to 24**. Crucially, **all 18 endpoint routes and operation IDs remain 100% identical and intact**. This document demonstrates every exact change.

1. High-Level Contract Diff Matrix

Contract Section	Main Branch (newcp.new.openapi.yaml)	Canonical (newcp.openapi.yaml)	Nature of Change
Endpoints & Paths (18 operations)	18 paths defined across auth, IAM, lookups, catalog, orders, and services.	Exact same 18 paths intact with zero path, parameter, or HTTP method removals.	**Identical (0 diffs)**
FormAnswer Schema Strictness	additionalProperties unconstrained (implicitly true). Allowed arbitrary properties.	additionalProperties: false strictly enforced. Prevents unexpected payload junk.	**Breaking:** Rejects unknown fields with 400 Bad Request.
FormAnswer Legacy Carriers	Contained 5 carrier fields: optionCode, valueText, valueNumber, valueBoolean, valueDate.	**All 5 carrier fields deleted.** Unified to optionCodes (array) and value (polymorphic).	**Breaking:** Eliminates carrier field clutter.
ServiceItem Required Fields	Only 8 properties marked required. Critical DB columns (like quantity, lineNumber) were optional.	**24 properties marked required**, matching V3 DB constraints and backend aggregate DTO guarantees.	**Type Refinement:** Eliminates defensive ? chaining in SDK.
Financial & Invoiced Locks	Generic note on financial locks.	Explicit documentation that invoiced lines reject changes to unitPrice, quantity, currency, and payer.	**Documentation Clarity**
Billed Party Route Description	Unquoted multiline description causing YAML scanner error in strict parsers.	Wrapped in quotes and properly indented. Resolves parser error.	**Syntax Fix**

2. Deep Dive: FormAnswer Schema Diff

The most critical architectural change between the main branch contract and the canonical contract is the complete pruning of dual-mode compatibility carriers from FormAnswer. Below is the exact side-by-side YAML diff:

MAIN BRANCH: newcp.new.openapi.yaml (DRAFT)

```yaml
FormAnswer:
  type: object
  # additionalProperties: not set
  required:
    - fieldCode
  properties:
    fieldCode:
      type: string
    value:
      description: Natural polymorphic answer value
      oneOf:
        - type: string
        - type: number
        - type: boolean
    optionCodes:
      type: array
      items:
        type: string
    # LEGACY CARRIERS ON MAIN BRANCH:
    optionCode:
      type: string
    valueText:
      type: string
    valueNumber:
      type: number
    valueBoolean:
      type: boolean
    valueDate:
      type: string
      format: date
  description: Form answer model supporting both...
```

CANONICAL: newcp.openapi.yaml (AFTER CHANGES)

```yaml
FormAnswer:
  type: object
  additionalProperties: false  # STRICT
  required:
    - fieldCode
  properties:
    fieldCode:
      type: string
      example: arch_selection
    value:
      oneOf:
        - type: string
        - type: number
        - type: boolean
      description: Natural polymorphic value...
      example: Lower
    optionCodes:
      type: array
      items:
        type: string
      description: Selected option codes...
      example:
        - lower
        - pontic_mandibular
    # ALL 5 CARRIERS PERMANENTLY REMOVED!
  description: Form answer model supporting unified
    optionCodes array and polymorphic value.
```

Detailed Breakdown of Removed vs Retained Properties

Property	Status in Canonical	Reason for Removal / Retention	Replacement Mechanism
`optionCode`	**REMOVED**	Redundant singular form. Created conflicting client code paths where both optionCode and optionCodes were submitted.	Use optionCodes: ['val'] (array of 1 item for single-select).
`valueText`	**REMOVED**	Legacy carrier from pre-canonical client. Unnecessary redundancy when polymorphic value supports strings naturally.	Use value: 'text here'.
`valueNumber`	**REMOVED**	Legacy numeric carrier. Caused type casting ambiguity when clients sent stringified numbers.	Use value: 3.5 (native JSON number).
`valueBoolean`	**REMOVED**	Legacy boolean carrier.	Use value: true (native JSON boolean).
`valueDate`	**REMOVED**	Legacy date carrier. Handled by backend string validation against ISO 8601 YYYY-MM-DD pattern.	Use value: '2026-11-05' (ISO string).
`fieldCode`	**RETAINED (Req)**	Primary identifier referencing catalog.fields.code.	Unchanged.
`optionCodes`	**RETAINED**	Unified array handling both single-select (0..1 items) and multi-select (0..n items).	Fixes the multi-select bug where only first option was persisted.
`value`	**RETAINED**	Polymorphic oneOf [string, number, boolean].	Standard natural representation for all free-form fields.

3. ServiceItem Schema: Expansion to 24 Required Fields

On the main branch, ServiceItem.required only listed 8 fields, meaning generated TypeScript clients treated 16 core database columns as optional (e.g. quantity?: number, lineNumber?: number). In newcp.openapi.yaml, all 24 columns guaranteed by the PostgreSQL V3 database and backend aggregate mappers are required.

MAIN BRANCH: ServiceItem.required (8 fields)

```
ServiceItem:
  type: object
  required:
    - uuid
    - orderUuid
    - sellableItemUuid
    - status
    - answers
    - billedToContactUuid
    - billedPartyOrigin
    - isBillable
  # (ONLY 8 FIELDS REQUIRED ON MAIN)
```

CANONICAL: ServiceItem.required (24 fields)

```
ServiceItem:
  type: object
  required:
    - uuid                - quantity
    - orderUuid           - isBillable
    - sellableItemUuid    - billedToContactUuid
    - lineNumber          - billedPartyOrigin
    - status              - canEditBilledParty
    - isRedo              - canEditPrice
    - isInvoiced          - canEditBillability
    - currencyCode        - answers
    - unitPrice           - etag
    - fieldsAmount        - createdAt
    - discountAmount      - updatedAt
    - voucherAmount
    - lineTotal
  # (ALL 24 FIELDS REQUIRED IN CANONICAL)
```

Inventory of the 16 Newly Guaranteed Properties

Newly Guaranteed Field	Type & Format	PostgreSQL V3 Database Backing	SDK & UI Benefit
`lineNumber`	integer	orders.order_items.line_number NOT NULL	Guarantees item ordering; no fallback index required.
`isRedo`	boolean	orders.order_items.is_redo NOT NULL DEFAULT false	Clean boolean for redo badge display.
`isInvoiced`	boolean	orders.order_items.is_invoiced NOT NULL DEFAULT false	Global lock trigger for financial edit controls.
`currencyCode`	string (ISO 4217)	orders.order_items.currency_code NOT NULL	Reliable currency formatting in monetary components.
`unitPrice`	number (2 decimals)	orders.order_items.unit_price NOT NULL	Guaranteed pricing breakdown without NaN fallbacks.
`fieldsAmount`	number (2 decimals)	orders.order_items.fields_amount NOT NULL	Reflects custom field add-on charges.
`discountAmount`	number (2 decimals)	orders.order_items.discount_amount NOT NULL	Item-level promotional discount representation.
`voucherAmount`	number (2 decimals)	orders.order_items.voucher_amount NOT NULL	Item-level voucher discount representation.
`lineTotal`	number (2 decimals)	orders.order_items.line_total NOT NULL	Computed final line price.
`quantity`	number (>= 0.001)	orders.order_items.quantity NOT NULL	Fractional quantity; removes item.quantity ?? 1 defensive code.
`canEditBilledParty`	boolean	Server permission & invoice lock aggregate evaluation	Directly binds to disabled state of Bill-To dropdown.
`canEditPrice`	boolean	Staff capability & line status lock evaluation	Directly binds to disabled state of unit price input.
`canEditBillability`	boolean	Staff capability & billing status lock evaluation	Directly binds to disabled state of billable toggle switch.
`etag`	string	Hash of item fields, answers, and modification timestamp	Guarantees optimistic concurrency for PATCH / transitions.
`createdAt, updatedAt`	string (date-time)	orders.order_items.created_at / updated_at NOT NULL	Accurate audit timestamps for timeline rendering.

4. Contextual & Operational Contract Clarifications

In addition to schema modifications, the canonical contract incorporates key documentation and description refinements that clarify server-side enforcement rules and prevent client parsing issues:

Target Location in Contract	Main Branch Text (newcp.new)	Canonical Contract Text (newcp.openapi)	Impact / Significance
/eligible-billed-parties (line 584)	Unquoted multiline description block indented at same column as key.	Explicitly single-quoted multiline description block with correct indentation.	Fixes yaml.scanner.ScannerError in strict YAML parsers (e.g. PyYAML / Spectral).
CreateServiceItem.answers (line 1253)	No description provided on the answers array property.	"Submitted form answers. One entry per answered field; see FormAnswer."	Clarifies that clients should only submit one answer object per field code.
PatchServiceItem.answers (line 1312)	Generic description on answers array.	"Complete replacement list of form answers. One entry per answered field; see FormAnswer."	**Critical semantic rule:** PATCH does not merge answers; it overwrites the full set.
PatchServiceItem (line 1361)	General note: "Other monetary changes obey their own permissions and V3 financial locks."	Added explicit specifics: "(e.g. invoiced lines reject unitPrice, quantity, currency, and payer changes)".	Documents exact HTTP 422 rejection behavior for invoiced service items.
ServiceItem.answers (line 1444)	No description provided on the response array.	"Form answers for this service instance. One entry per answered field; see FormAnswer."	Confirms response aggregation matches submission structure.

5. Architectural Impact & System Alignment

WHY THESE CHANGES BENEFIT THE ENTIRE PLATFORM:

- **Zero Ambiguity in Payload Serialization:** Eliminates confusing heuristics where backend mappers had to guess whether to read optionCode, optionCodes[0], or valueText.
- **Fixes Multi-Select Bug Permanently:** The main branch implementation suffered from a defect where only optionCodes[0] was saved to the database. The canonical schema and model enforce one database row per selected option code, properly restoring multi-select answers on read.
- **Complete Alignment with PostgreSQL V3:** Database columns with NOT NULL constraints are now mirrored accurately in the OpenAPI specification, preventing runtime discrepancies between Drizzle ORM models and frontend types.
- **Predictable Client-Side UI States:** With canEditBilledParty, canEditPrice, and canEditBillability guaranteed present, frontend components can deterministically disable inputs without calculating complex invoice and permission states on the client.

6. Summary Sign-Off

Review Criterion	Verification Status	Notes
Endpoint Preservation	**VERIFIED - ALL 18 PATHS INTACT**	No endpoints, methods, or path parameters were altered or dropped.
OpenAPI 3.0.3 Syntax	**VALID - 0 SYNTAX ERRORS**	Passed full parsing validation with js-yaml and PyYAML; all $refs resolve.
Backend Implementation Status	**CONTRACT READY**	Served via openapi/newcp.openapi.yaml by src/api-docs.ts.